LINUX SYSTEM ADMINISTRATION • GUIDED EXERCISE

Regular Expressions & Pattern Matching in Linux

Searching • Filtering • Validating • Extracting text

```
$ grep -E '^A' students.txt
Amit CSE
```

^ $. * + ?

[ ] { } |

TOOLS

grep **grep -E** **sed** **awk**

What Is Pattern Matching?

Pattern matching means searching text for characters or sequences that satisfy a specified pattern.

EXAMPLE FILE: students.txt

Amit CSE
Rahul CSE
Neha ECE
Priya CSE
Rohan ME

COMMAND

```
$ grep "CSE" students.txt
```

OUTPUT

```
Amit CSE
Rahul CSE
Priya CSE
```

3 of 5 lines contain “CSE”

What Is a Regular Expression?

A regular expression (regex) is a special pattern used to describe text.

SIMPLE SEARCH

grep "CSE" students.txt

REGEX SEARCH

grep -E "^A" students.txt

^A means: lines that start with A.

COMMAND

```
$ grep -E "^A" students.txt
```

OUTPUT

```
Amit CSE
```

PATTERN BREAKDOWN

^	start of line
A	the letter A

CORE SYMBOLS

Basic Regex Symbols

^ Start of line	$ End of line	. Any single character	[abc] a, b, or c
[^0-9] Not a digit	[0-9] Any digit	+ One or more	* Zero or more
? Zero or one	{n} Exactly n	{n,m} n to m occurrences	| OR

^ and $ — Start and End of Line

^

^app → line must begin with app

COMMAND

```
$ grep '^app' fruits.txt
```

OUTPUT

```
apple
application
```

$

$ → line must end with the specified pattern

COMMAND

```
$ grep 'CSE$' students.txt
```

OUTPUT

```
Amit CSE
Rahul CSE
Priya CSE
```

. and [ ] — Character Matching

. = any one character

COMMAND

```
$ grep -E 'c.t' file.txt
```

OUTPUT

```
cat
cut
cot
c9t
```

[ ] [0-9] = any digit from 0 to 9

COMMAND

```
$ grep -E '[0-9]' file.txt
```

CHARACTER SETS

[0-9] digit
[a-z] lowercase letter
[A-Z] uppercase letter
[^0-9] not a digit

Quantifiers: *, + and ?

*

ab*

a followed by zero or more b's

+

ab+

a followed by one or more b's

?

colou?r

u is optional

COMMAND

```
$ grep -E 'ab+' file.txt
```

OUTPUT

```
ab
abb
abbb
```

Use grep -E for extended operators such as + and ?.

{n}, {n,} and {n,m} — Repetition

{n}

^[0-9]{5}$

exactly 5 digits

{n,}

^[0-9]{3,}$

3 or more digits

{n,m}

^[0-9]{3,5}$

3 to 5 digits

COMMAND

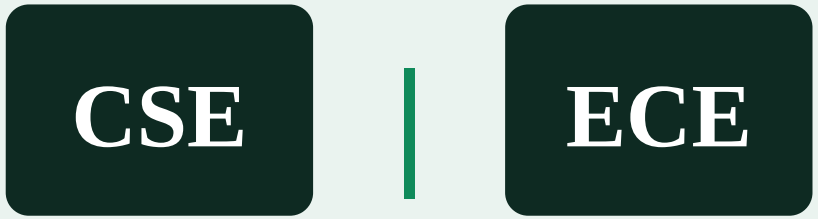
```
$ grep -E '^[0-9]{5}$' data.txt
```

OUTPUT

```
12345
98765
```

^ and $ anchor the pattern, so the whole line must match.

| — OR Operator



CSE|ECE means: CSE OR ECE

Useful for filtering multiple alternatives.

COMMAND

```
$ grep -E 'CSE|ECE' students.txt
```

OUTPUT

```
Amit CSE
Rahul CSE
Neha ECE
Priya CSE
```

APPLICATIONS

Email Validation with Regex

USERNAME	@	DOMAIN	.	EXTENSION
^[A-Za-z0-9._%+-]+	@	[A-Za-z0-9.-]+	\.	[A-Za-z]{2,}$

COMMAND

```
$ grep -E '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$' emails.txt
```

OUTPUT

```
student@gmail.com
teacher@lpu.in
admin@company.org
john.doe@gmail.com
```

Regex in a Shell Script

SCRIPT

```
#!/bin/bash
echo "Enter email:"
read email

if echo "$email" | grep -Eq '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$'
then
    echo "Valid email"
else
    echo "Invalid email"
fi
```

RUN 1

```
Enter email:
student@gmail.com
Valid email
```

RUN 2

```
Enter email:
student@gmail
Invalid email
```

grep vs grep -E

grep

BRE

- Basic Regular Expressions (BRE)
- Good for simple searches and basic patterns

EXAMPLE

```
$ grep "error" logfile.txt
```

grep -E

ERE

- Extended Regular Expressions (ERE)
- Easier use of +, ?, |, {n,m}

EXAMPLE

```
$ grep -E 'ERROR|WARNING' logfile.txt
```

APPLICATIONS

Why Regex Is Important in Linux

01

Search text in files

02

Filter command output

03

Validate email, phone, username, etc.

04

Analyze system/application logs

05

Extract useful information

06

Build decision-making shell scripts

COMMON TOOLS **grep • sed • awk**

PRACTICE

Practice Tasks

1. Find lines beginning with a vowel.
2. Find lines ending with CSE.
3. Find names containing exactly five characters.
4. Find numbers containing exactly four digits.
5. Find numbers beginning with 9.
6. Find lines containing CSE or ECE.
7. Validate a phone number using regex.
8. Validate an email using regex.
9. Check a username containing only letters and digits.
10. Write a shell script for email validation.

PRACTICE

Quick Revision

`^abc`	starts with abc	`[0-9]{5}`	exactly 5 digits
`abc$`	ends with abc	`a+`	one or more a
`[0-9]`	digit	`a*`	zero or more a
`[A-Z]`	uppercase letter	`a?`	zero or one a
`[a-z]`	lowercase letter	`cat|dog`	cat OR dog

Remember: Regex describes the pattern; grep applies that pattern to text.